D_QCOH HAS DIRECT SUMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 567-577, immediately following the statement at 561-565.

Editorial note. The preceding lemma in the source (quasi-coherence of E itself) ends with a pure-citation proof and is not part of this span.

Allowed standard prerequisites (editorial).

- Direct sums in the derived category. Injectives, Lemma injectives-lemma-derived-products, cited at proof lines 568-569. D(O_X) has direct sums, computed termwise on representatives.
- Quasi-coherent sheaves are closed under direct sums. Properties of Spaces, Lemma spaces-properties-lemma-properties-quasi-coherent, cited at proof lines 575-577. Terminal input: termwise direct sums of quasi-coherent sheaves are quasi-coherent.

Source reference. Original reference 07D9: focused-seeds/geometry/S9-perf-quasi-coherence-direct-sums/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M1: focused-seeds/geometry/S9-perf-quasi-coherence-direct-sums/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Then $D_{QCoh}(\mathcal{O}_X)$ has direct sums.*

Proof. By Injectives, Lemma 07D9 the derived category $D(\mathcal{O}_X)$ has direct sums and they are computed by taking termwise direct sums of any representatives. Thus it is clear that the cohomology sheaf of a direct sum is the direct sum of the cohomology sheaves as taking direct sums is an exact functor (in any Grothendieck abelian category). The lemma follows as the direct sum of quasi-coherent sheaves is quasi-coherent, see Properties of Spaces, Lemma 03M1. □

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.